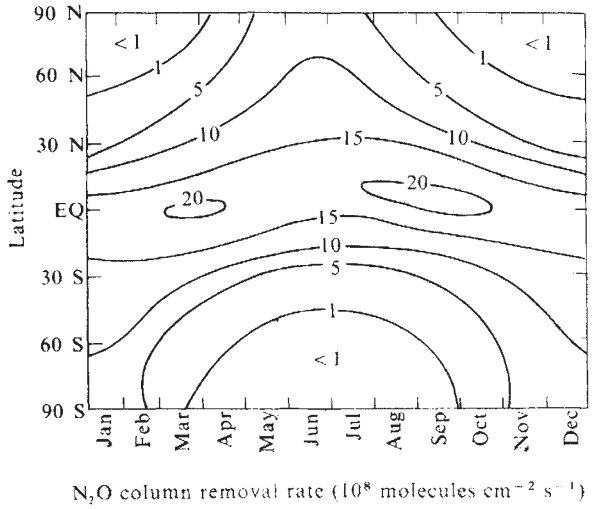


Fig. 3 Calculated CFC-11 column removal rate as a function of latitude and time of the year. The model simulation was performed using a boundary value of 1 p.p.b.v. at the ground.

We have also calculated the lifetimes for CFC-11 and CFC-12 using a steady-state simulation. Direct comparison of the calculated gas distributions of this model simulation with observations is not possible as the gases have not attained the steady-state concentrations in the present atmosphere. As an alternative, the computed CFC-11 and CFC-12 distributions corresponding to the year 1979 of a model simulation using known release rates[17] can be compared with the available observations[8–14]. Agreement is reasonably good for CFC-11 and CFC-12 at the Equator but poor at mid-latitudes[18]. The calculated mid-latitude CFC-11 and CFC-12 mixing ratios above 25 km are higher than the observations by about a factor of 2–4. The discrepancy could be due to either the error in the photolysis rates or error in the transport coefficients. The column removal rate of CFC-12 has features very similar to that of N_2O (Fig. 2). For CFC-11, the latitudinal gradient of the removal rate is stronger as shown in Fig. 3. The calculated lifetimes are 65 and 135 yr for CFC-11 and CFC-12 respectively.

The corresponding 2-D model calculated lifetimes reported by Hinds[19] are 81 and 182 yr and those from the Oxford University model[3] are 78 and 133 yr respectively. Direct comparison of these values with our calculated values is difficult because it is not clear whether the values reported correspond to steady-state lifetimes.

Note that the calculated lifetimes given here also depend on model structure, such as parameterization for transport and treatment of photochemistry. The question of whether these calculated values truly represent the actual atmospheric lifetimes can be answered only following further validation of the corresponding calculated distributions using more observational data and improvement in transport and chemical data. However, we believe that the latitudinal behaviour of the column removal rate simulated by the 2-D model is qualitatively correct and that the positive correlation between the latitudinal variation of f and L should not be ignored.

Although 1-D models are still the most widely used diagnostic and prognostic tools in stratospheric research, two limitations should be noted with regard to their lifetime predictions: first, their inability to incorporate the covariance effects discussed here; second, and more problematic, the inherent difficulty in validating their lifetime predictions using field data. Our discussion points to the fact that, even for a fully validated 1-D model in which the calculated profiles agree with global average observations, the model may still underestimate the global removal rate and, therefore, overestimate the lifetime because of its inability to account for the positive contribution from the covariance of f and L. With a modest increase in computer time, 2-D models, on the other hand, can simulate the latitudinal and seasonal variations of atmospheric species as well as the solar insolation and may therefore be useful in refining the lifetime estimates calculated by 1-D models. In light of the above discussion, a combination of selective field data and a validated 2-D model appears to be the best approach to obtaining more accurate lifetime estimates. Obviously, 2-D models are still in a developmental state, and more atmospheric data are needed for further validation and improvement.

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Position of the Lhasa block, South Tibet, during the late Cretaceous

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The latitude of the Lhasa block (Xizang, South Tibet) during the late Cretaceous–early Tertiary is of prime importance for understanding the collision of India and Eurasia and the mechanism of the formation of the Qinghai–Xizang plateau. During the first field expedition of the French–Chinese collaboration, several geological formations have been sampled in southern Tibet between Lhasa (29.40° N; 91.09° E) and Dingri (28.35° N; 86.38° E). We present here the first results obtained for middle Cretaceous to Palaeocene red beds from the Lhasa block, north of the Yarlung–Zang Bo suture zone, recognized as the main suture between the Indian and the Eurasian plates. They indicate a palaeolatitude of the Lhasa block of 20° N during the late Cretaceous while India was situated 30° S.

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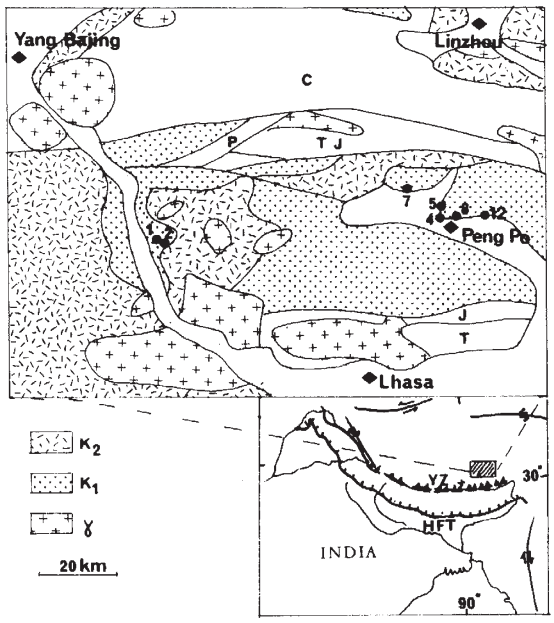


Fig. 1 Geological outlines of the Lhasa region: C, Carboniferous; P, Permian; T, Trias; J, Jurassic. K_1, Takena Formation and lower-middle Cretaceous; K_2, Lingzizong Formation; γ, granitic intrusives. Redrawn from geological map of Lhasa–Nyalam Area (Academia Sinica). YZ, Yarlung Zang Bo suture zone; HFT, Himalayan Frontal Thrust.

The geological formations that we sampled were red beds from Linzhou basin. Continental sediments are present all over the Tibetan plateau and particularly in the Linzhou basin northeast of Lhasa (Fig. 1) where they are represented by two formations separated by a major unconformity. The rock lying under the unconformity is Takena Formation of middle to late Cretaceous age, mainly consisting of sandstone, limestone and shale, strongly folded and with a total thickness of ~1 km. Volcanics (andesite and dacite) are interbedded at the base of this formation. Above the unconformity, Lingzizong Formation of late Cretaceous age mainly consists of black shale in its lower part and red sandstone often conglomerate upwards. A thick sequence of greyish-blue andesite overlays the Lingzizong Formation. Red sandstones of the Takena Formation were sampled (Fig. 1) at sites 1, 2, along the road Lhasa–Yangbajin and at sites 4, 5, 7, 8, 12 near Linzhou (Peng Po Farm).

Samples were taken with a portable drill or as block samples, and were oriented with magnetic and solar compass. Individual sites were sampled laterally over 50–100 m and vertically over 10–20 m.

The measurements were done in Paris and Strasbourg on standard 25-mm cores using either a Schönstedt or a Digico Spinner magnetometer. The two series of measures were cross checked. Samples were demagnetized by step heating and cooling in a zero field (better than 20 nT). The different magnetic components were analysed using the classical Zijderveld vector diagram method.

Thermal cleaning shows the presence of two magnetic components in approximately half of the samples. One is eliminated after heating at 400–500 °C and does not show a clear mean direction. The other is only broken down above 650 °C and is usually the last component. The directions of this component are well grouped. Other samples show only one direction eliminated above 650 °C. Before tectonic correction, the mean directions are generally different from the present field.

Table 1 shows the results before and after tectonic corrections. k increases from 7.3 to 35 by a factor of 4.8. Statistical tests have been used to estimate the validity of the fold test. McElhinny's test[1], which is very stringent, is positive at 95% significance level. McFadden and Jones[2] devised another test but this was difficult to use when each site had a different tectonic setting. We can confidently conclude that the fold test is positive and that the magnetization predates the late Cretaceous–Palaeocene folding phase.

In site 7, four specimens are of normal polarity while seven are inverse, the two directions being antiparallel. The presence of normal and reverse polarity in site 7 of red and green sandstone also argue for a primary origin of the hard component of magnetization.

Nevertheless, against the strong technical arguments for primary origin, one has to consider (Fig. 2) that before bedding correction, the virtual geomagnetic poles (VGP) for sites 1, 2, 4, 5, 8 and 12 are not significantly different (C. T. Klootwijk, personal communication) from the apparent polar wander path (APWP) for India. In the Linzhou basin, tectonic movements of comparatively little amplitude occurred after the main late Cretaceous–early Tertiary folding phase. VGP from site 7 with

Table 1 Palaeogeomagnetic results before and after tectonic correction

Site	Before unfolding N	D	I	After unfolding D	I	k	$\alpha 95$
Present results							
1	8	335	2	324	37	68	7°
2	7	341	9	333	43	16	16°
4	9	350	42	333	28	90	5°
5	10	352	33	328	47	36	8°
7*	11	10	25	3	24	69	6°
8	11	0	52	338	29	51	6°
12	12	344	−33	341	42	48	6°
Mean before tectonic correction		350	20	—	—	7, 3	24
After tectonic correction		—	—	338	36	35	10
Mean PGV: $N = 7$ sites Lat. 68° N Long. 340° E Palaeolatitude for Lhasa 20° N							
Previous results[3,4]							
Red beds (Linzhou)							
Zhu Xiang Yuan	12	—	—	335	16	66	5°
Zhu Zhi Wen	42	—	—	338	40	20	7°
Lhasa Upper Jurassic	21	—	—	175	2	5	17°

* Both reverse and normal samples.

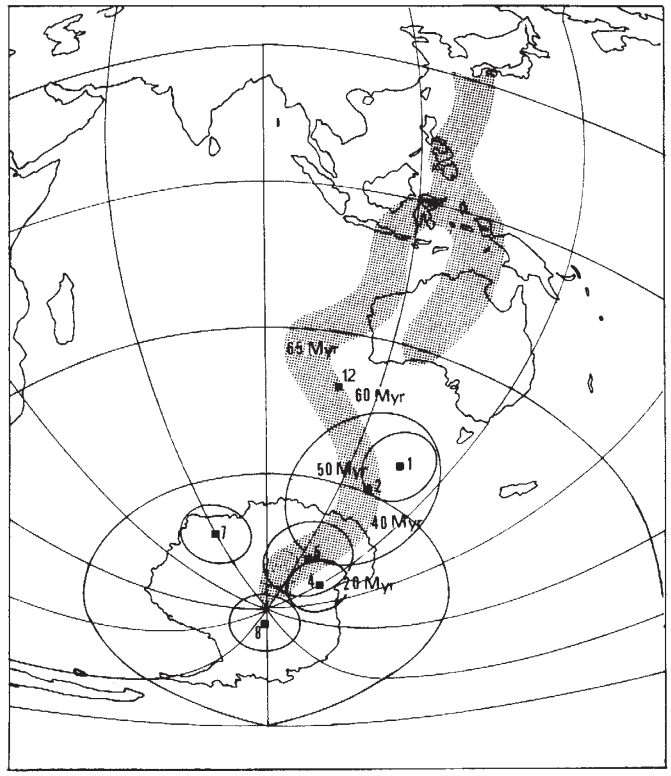


Fig. 2 APWP for India (from ref. 6) compared with the geomagnetic poles of Cretaceous Tibetan sediments and volcanics before unfolding.

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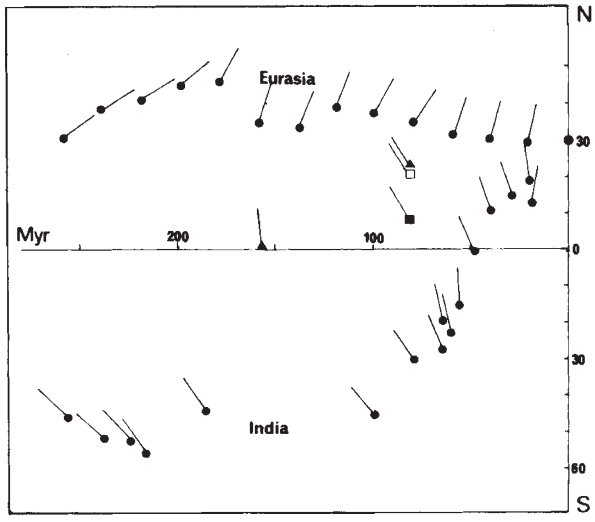


Fig. 3 Palaeolatitude and palaeoorientation of a point now in Lhasa as if it were bound to Eurasia (upper curve), or as if it were bound to India (lower curve). Recalculated from ref. 5 and from refs 6, 7. □, Results from this paper; ▲, results from ref. 4; ■, results from ref. 3.

both normal and reverse layers is different from the APWP of India. Although only a few sites are considered here, it seems that VGPs from site 12, from sites 1 and 2 and from sites 4 and 5 correspond respectively to periods of 65, 50 and 25 Myr on the APWP for India, the two last being critical periods for remagnetization. This could argue for a remagnetization of these formations, when the Lhasa block collided with India. If remagnetization occurred, other methods of investigations will be necessary to decipher the magnetization, or other types of formations should be used to assess the palaeolatitude of Lhasa block; useful conclusions can then be deduced from overprinted components concerning the latitude of the collisions, but it will be difficult to explain how three phases of remagnetization have affected, first site 12, then sites 1 and 2 and possibly sites 4, 5 at 65–50 Myr in a near equatorial latitude, and finally at higher latitude site 4, at 25 Myr.

Until now, the positive fold test has led us to conclude that a complete remagnetization probably did not occur and that the results give a good approximation of the position of the Lhasa block during the Cretaceous.

Chinese palaeomagneticists have done several studies in Xizang, only a few of which have been published. Zhu Xiang Yuan *et al.*[3] have sampled the Takena red beds probably not far from site 4 (the Yayong hill near Peng Po farm). A few samples were a.f. demagnetized up to 30 mT and do not show any directional change. The result given is based on natural remanent magnetism measurements and is dip corrected.

A more extensive study has been published by Zhu Zhi Wen *et al.*[4] over Xizang. Results concerning the Lhasa block are on red beds from Takena (Lhasa cement mill). All samples were a.f. demagnetized up to 50 mT and the results corrected for tilt are given in Table 1.

Compared with the Chinese results for the Linzhou red beds, new results have been obtained on more widely distributed sites using thermal demagnetization techniques which are generally more effective for resolving multi-component magnetizations of red beds than are a.f. techniques. Indeed, a.f. demagnetizations performed on red beds by Zhu Zhi Wen and others leave 90% of the initial magnetization undestroyed at the peak field used (50 mT).

Figure 3 gives the palaeolatitude of Eurasia recalculated from ref. 5 and India from refs 6, 7 as a function of time, with the results of Zhu Zhi Wen[4] for Jurassic limestone and Cretaceous red beds. Arrows indicate declinations with respect to the north and show an anticlockwise rotation of the Lhasa block since the Jurassic. During the middle Cretaceous, the palaeolatitude of the Lhasa block was about 20° N. The results of Zhu Zhi Wen[4] indicate a near equatorial latitude during the Jurassic. These results confirm that the Lhasa block, which is of Gondwanan origin and has preceded northwards drifting India, was widely separated from India during the late Cretaceous and possibly during the Jurassic according to Zhu Zhi Wen results (Fig. 3). During the Cretaceous the northern edge of India was situated near 30° S and the southern margin of Asia probably not far from 30° N. As the palaeolatitude of the Lhasa block was about 20° N it is likely that the Tertiary convergence between South Tibet and India was much larger due to subduction and crustal shortening[8] than between South Tibet and Asia. The anticlockwise rotation of the Lhasa block (Fig. 3) probably occurred under the constraint of the anticlockwise rotation of northward moving India block.

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Palaeoclimates at Lake Turkana, Kenya, from oxygen isotope ratios of gastropod shells

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Climatic changes in the Plio-Pleistocene of East Africa during the evolutionary development of early man have been identified only in their broadest outlines. A variety of palaeontological[1–3] and geochemical[4] techniques have already been applied to the study of palaeoclimates in this area, but the use of stable oxygen isotope ratios, so helpful in marine palaeoclimatology, has not been fully exploited[5,6]. The frequent occurrence of well-preserved mollusc shells in the sedimentary column at Lake Turkana provides a sensitive oxygen isotope record, varying over a range of 7‰, which can be interpreted in terms of evaporative concentration of the heavier oxygen isotope in waters of the closed lake, alternating with isotopically lighter freshwater when the lake has been open. I report here the results of a study of the $^{18}O/^{16}O$ ratios of three species of gastropods from the sediments of the Koobi Fora region of Lake Turkana, and evaluate how the technique might be applied to the study of the palaeoclimates of other tropical lakes.

The sedimentary deposits of palaeo-Lake Turkana in northern Kenya are well known for their yield of hominid fossils[7] and artefacts[8] of early man. They also include numerous and varied land and freshwater fossils, including the frequent occurrence of well-preserved mollusc shells. A well-documented collection of these shells[9] provided the impetus and raw materials for this study. Among the 95 shell-bearing horizons recognized by Williamson in his study of the palaeontology of the Lake Turkana molluscs, some 36 contain the gastropods *Mellanoides tuberculata*, *Cleopatra ferruginea* and *Bellamya unicolor*. These three were chosen from among the ~20 species identified by Williamson because: (1) they occur in shallow water habitats and would be expected to be in good equilibrium with the surrounding water; (2) they are capable of living over a wide range of *p*H and concentrations of dissolved salts

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